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Assessment I - CO1

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DARSHAN D

1. A hospital dashboard displays patient age using point size and blood pressure using point color. Elderly patients appear larger, while blood pressure is represented using a blue-to-red gradient. Which visualization principle is primarily violated?

*

- ☐ A. Position should not encode quantitative values.
- ☒ B. Point size is unsuitable for ordered quantitative comparisons.
- ☐ C. Color gradients cannot represent quantitative values.
- ☐ D. Age should always be mapped to the x-axis.

2. A financial analyst uses different shapes to represent monthly stock prices over time. Users struggle to identify trends.

Which mapping would improve quantitative interpretation? *

- ☐ A. Map stock prices to shape.
- ☐ B. Map stock prices to color hue.
- ☒ C. Map stock prices to vertical position.
- ☐ D. Map stock prices to transparency.

3. A researcher maps customer satisfaction levels (Very Poor, Poor, Fair, Good, Excellent) to random colors without ordering.

The visualization becomes difficult to interpret because

*

- ☐ A. Satisfaction is nominal.
- ☒ B. Satisfaction is ordinal and requires ordered aesthetics.
- ☐ C. Random colors increase contrast.
- ☐ D. Color hue automatically implies ordering.

4. A climatologist maps rainfall intensity simultaneously to point size and color intensity.

The visualization becomes misleading because

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- ☒ A. Redundant encoding may exaggerate perceived differences.
- ☐ B. Color cannot represent quantitative variables.
- ☐ C. Point size cannot encode quantitative data.
- ☐ D. Multiple aesthetics always improve interpretation.

5. An analyst represents employee departments using a continuous color gradient.

Which issue arises?

*

- ☒ A. Continuous scales imply ordering where none exists.
- ☐ B. Gradients improve categorical distinction.
- ☐ C. Departments are quantitative.
- ☐ D. Cartesian coordinates become invalid.

6. A scatter plot maps income to x-position and education level to y-position. Gender is represented using color.

Which aesthetic represents nominal data?

*

- ☐ A. x-position
- ☐ B. y-position
- ☒ C. Color

☐ D. Position

7. A visualization maps annual sales to point transparency instead of position.

Why is this a poor choice?

*

- ☒ A. Transparency has low perceptual accuracy.
- ☐ B. Transparency cannot encode variables.
- ☐ C. Transparency represents categories only.
- ☐ D. Position should only represent nominal data.

8. A chart uses texture patterns instead of colors to distinguish land-use categories.

The primary advantage is

*

- ☐ A. Improved quantitative precision
- ☒ B. Better accessibility for color-blind users
- ☐ C. Reduced memory usage
- ☐ D. Faster rendering speedtion

9. A dataset contains Country, GDP, Population, and Continent.

Which variable should ideally be encoded using shape?

*

- ☐ A. GDP
- ☐ B. Population
- ☒ C. Continent

☐ D. GDP Growth

10.

Which visualization mapping produces the highest perceptual accuracy for comparing numerical values?

*

- ☐ A. Area
- ☐ B. Color satur
- ☒ C. Position along a common scale
- ☐ D. Transparency

11. A data scientist uses a logarithmic scale for household income.

Which reason best justifies this choice?

*

- ☐ A. Income contains many categories.
- ☒ B. Income spans several orders of magnitude.
- ☐ C. Log scales improve categorical comparis
- ☐ D. Logarithmic scales remove outliers.

12. A temperature visualization assigns identical colors to values between 20°C and 25°C.

The scale primarily reduces

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- ☒ A. Precision
- ☐ B. Coordinate transformation
- ☐ C. Scatter density

☐ D. Position accuracy

13. A continuous variable is mapped to five discrete colors.

Which problem is introduced?

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- ☐ A. False precision
- ☒ B. Information loss
- ☐ C. Axis inversion
- ☐ D. Overplotting

14. A scatter plot uses identical symbol sizes for all observations while mapping sales to color intensity.

Which aesthetic is unused?

*

- ☐ A. Position
- ☒ B. Size
- ☐ C. Color
- ☐ D. Shape

15.

When transforming data values into visual positions, the mapping function is called

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- ☐ A. Coordinate system
- ☒ B. Scale

- ☐ C. Axis
- ☐ D. Legend

16. A scientist rotates Cartesian coordinates by 45° .

Which property remains unchanged?

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- ☒ A. Relative distances
- ☐ B. Scale labels
- ☐ C. Color palette
- ☐ D. Shape encoding

17. A visualization requires comparison of radial distances from a central point.

Which coordinate system is appropriate?

*

- ☐ A. Cartesian
- ☒ B. Polar
- ☐ C. Parallel
- ☐ D. Ternary

18. A business analyst compares quarterly revenues using pie slices arranged radially.

Which issue is most likely?

*

- ☒ A. Difficulty comparing angles accurately
- ☐ B. Cartesian distortion
- ☐ C. Incorrect logarithmic transformation
- ☐ D. Missing legends

19. A map uses latitude and longitude.

These coordinates are examples of

*

- ☐ A. Polar
- ☐ B. Cartesian
- ☒ C. Geographic coordinate system
- ☐ D. Logarithmic coordinates

20. Why are Cartesian coordinates preferred for scatter plots? *

- ☐ A. They improve color perception.
- ☒ B. They preserve independent quantitative dimensions.
- ☐ C. They require fewer data points.
- ☐ D. They eliminate outliers.

21. A microbiologist visualizes bacterial counts ranging from 100 to 10 million.

Which axis is most suitable?

*

- ☐ A. Linear
- ☐ B. Polar

- ☒ C. Logarithmic
- ☐ D. Circular

22. Which statement about logarithmic axes is TRUE? *

- ☐ A. Equal distances represent equal differences.
- ☒ B. Equal distances represent equal ratios.
- ☐ C. Values cannot be negative.
- ☐ D. Both B and C.

23. A student uses a log axis for exam scores between 60 and 100.

The visualization becomes

*

- ☐ A. More informative
- ☒ B. Less meaningful because data span is small
- ☐ C. Easier to compare
- ☐ D. Better for categorical data

24. Applying nonlinear scaling mainly helps *

- ☐ A. Increase categorical distinction
- ☒ B. Compress skewed numerical distributions
- ☐ C. Improve color accessibility
- ☐ D. Eliminate missing values

25. Which transformation best linearizes exponential growth? *

- ☐ A. Square root
- ☒ B. Logarithm
- ☐ C. Reciprocal
- ☐ D. Mean normalization

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